

Seongmoon Jeong

PHD CANDIDATE

IRIS Lab., Sungkyunkwan University, Suwon, Republic of Korea

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I am a Ph.D. candidate in the Department of Artificial Intelligence at Sungkyunkwan University, advised by **Prof. Jong Hwan Ko**. My research focuses on **image compression for machine perception**, spanning low-level optimization and high-level tasks such as classification, object detection, and segmentation. I develop **learned, task-aware representations** that jointly optimize compression efficiency and downstream vision performance. More recently, I have investigated high-level vision directly in the **RAW domain**, before the image signal processing (ISP) pipeline, to preserve information relevant to machine perception. My long-term goal is to enable **communication-efficient intelligent systems** for autonomous systems and edge-based machine vision. **Available for full-time roles starting March 2027.**

Education

Ph.D. in Artificial Intelligence (Expected February 2027)

SUNGKYUNKWAN UNIVERSITY (SKKU)

Suwon, Korea

Mar. 2020 - Present

- Received an Outstanding Scholarship awarded to promising students in the Department of Artificial Intelligence.

B.S. in Electronic and Electrical Engineering

SUNGKYUNKWAN UNIVERSITY (SKKU)

Suwon, Korea

Mar. 2013 - Feb. 2020

- GPA: 3.84/4.5 (Top 6.6%)

Research Experience

Graduate Researcher

IRIS LAB, SUNGKYUNKWAN UNIVERSITY (SKKU)

Suwon, Korea

Mar. 2020 - Present

- Machine-Oriented Image Compression:** Codec-transferable and test-time adaptation across downstream vision tasks [**C4**, **C5**]
- Task-Aware Preprocessing and JPEG Coding:** Adaptive downscaling, feature modulation, and region-based JPEG optimization [**C1**, **C3**, **J1**]
- RAW-Domain Machine Vision:** Foundation-model-guided RGB-to-RAW generation for object detection [**C6**]

Project Experience

Video Coding for Machines

LEAD STUDENT RESEARCHER (IRIS LAB., SKKU)

Media Research Division, ETRI

2023 - 2024

Adaptive and Efficient Preprocessing and Coding for Machine Vision Tasks and Input Images

LEAD STUDENT RESEARCHER (IRIS LAB., SKKU)

Media Research Division, ETRI

2022

Selected Publications

C6 Foundation Model-Guided RGB-to-RAW Generation with Spectral Supervision for RAW-Domain Detection

Seongmoon Jeong, Yulhwa Kim, Jong Hwan Ko

IEEE AVSS

Sep. 2026

C5 UICAM: A Codec-Transferable Adapter for Machine-Oriented Image Compression

Seongmoon Jeong, Sangwoon Kwak, Soon-heung Jung, Hyon-Gon Choo, Jong Hwan Ko

IEEE AVSS

Sep. 2026

C4 Test-Time Fine-Tuning of Image Compression Models for Multi-Task Adaptability

Un Ki Park, Seongmoon Jeong, Youngchan Jang, Gyeongmoon Park, Jong Hwan Ko

IEEE CVPR

Mar. 2025

C3 Adaptive Image Downscaling for Rate-Accuracy-Latency Optimization of Task-Target Image Compression

Hangyul Choi, Seongmoon Jeong, Sangwoon Kwak, Soon-heung Jung, Jong Hwan Ko

IEEE AICAS

Feb. 2024

C1 Rate-Controllable and Target-Dependent JPEG-Based Image Compression Using Feature Modulation

Seongmoon Jeong, Kang Eun Jeon, Jong Hwan Ko

IEEE ICME Workshop

Apr. 2023

J1 An Overhead-Free Region-Based JPEG Framework for Task-Driven Image Compression

Seonghye Jeong, Seongmoon Jeong, Simon S. Woo, Jong Hwan Ko

Pattern Recognition Letters

Nov. 2022

Standardization Activities

[VCM] Neural network based post-filter for VCM

ISO/IEC JTC1 / SC29 / WG4 M71305

Geneva

Jan. 2025

[VCM] Neural network based pre-filter for VCM

ISO/IEC JTC1 / SC29 / WG4 M71304

Geneva

Jan. 2025

[VCM] Learned joint filter network for VCM

ISO/IEC JTC1 / SC29 / WG4 M69990

Kemer

Nov. 2024

[VCM] Learned pre-filter network for VCM

ISO/IEC JTC1 / SC29 / WG4 M69085

Sapporo

Jan. 2024

Honors & Awards

2024 **3rd Place**, DAC System Design Contest - GPU Track

San Francisco, USA

2020 **2nd Place**, Artificial Intelligence Grand Challenge, Ministry of Science and ICT

Korea

2019 **Honorable Mention**, Big Data Center Open Innovation Challenge

Seongnam, Korea

2019 **Outstanding Work Award**, Capstone Design Project, SKKU

Suwon, Korea

2018 **Dean's List Award**, Fall semester, SKKU

Suwon, Korea

2017 **Dean's List Award**, Fall semester, SKKU

Suwon, Korea

Full Publications

INTERNATIONAL CONFERENCE

C6 Foundation Model-Guided RGB-to-RAW Generation with Spectral Supervision for RAW-Domain Detection

IEEE AVSS

Seongmoon Jeong, Yulhwa Kim, Jong Hwan Ko

Sep. 2026

C5 UICAM: A Codec-Transferable Adapter for Machine-Oriented Image Compression

IEEE AVSS

Seongmoon Jeong, Sangwoon Kwak, Soon-heung Jung, Hyon-Gon Choo, Jong Hwan Ko

Sep. 2026

C4 Test-Time Fine-Tuning of Image Compression Models for Multi-Task Adaptability

IEEE CVPR

Un Ki Park, Seongmoon Jeong, Youngchan Jang, Gyeongmoon Park, Jong Hwan Ko

Mar. 2025

C3 Adaptive Image Downscaling for Rate-Accuracy-Latency Optimization of Task-Target Image Compression

IEEE AICAS

Hangyul Choi, Seongmoon Jeong, Sangwoon Kwak, Soon-heung Jung, Jong Hwan Ko

Feb. 2024

C2 Kernel Shape Control for Row-Efficient Convolution on Processing-In-Memory Arrays

IEEE ICCAD

Johnny Rhe, Kang Eun Jeon, Joo Chan Lee, Seongmoon Jeong, Jong Hwan Ko

Jul. 2023

C1 Rate-Controllable and Target-Dependent JPEG-Based Image Compression Using Feature Modulation

IEEE ICME Workshop

Seongmoon Jeong, Kang Eun Jeon, Jong Hwan Ko

Apr. 2023

INTERNATIONAL JOURNAL

J2 KERNTROL: Kernel Shape Control Toward Ultimate Memory Utilization for In-Memory Convolutional Weight Mapping

IEEE TCAS-I

Johnny Rhe, Kang Eun Jeon, Joo Chan Lee, Seongmoon Jeong, Jong Hwan Ko

Feb. 2024

J1 An Overhead-Free Region-Based JPEG Framework for Task-Driven Image Compression

Pattern Recognition Letters

Seonghye Jeong, Seongmoon Jeong, Simon S. Woo, Jong Hwan Ko

Nov. 2022

Student Mentoring

Hangyul Choi

M.S. STUDENT AT SUNGKYUNKWAN UNIVERSITY, NOW AT MX BUSINESS, SAMSUNG ELECTRONICS 2023 - 2024

- Adaptive Image Downscaling for Rate-Accuracy-Latency Optimization of Task-Target Image Compression

Chanung Park

M.S. STUDENT AT SUNGKYUNKWAN UNIVERSITY 2024

Skills

Programming Languages	Python, Shell scripting, C/C++
ML Frameworks	PyTorch, TensorFlow, JAX, TensorRT
Development Frameworks	Docker, Conda, uv

Reference

Jong Hwan Ko

ASSOCIATE PROFESSOR Ph.D. advisor

Dept. of ECE, SKKU, Suwon, Korea

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- jhko@skku.edu